

CC - Student Record Management System using Apex and Visualforce in Salesforce

Problem Statement

Develop an application in Salesforce using Apex programming language and Visualforce page to manage student records (Name, Roll No, Class, Mobile No) stored in a custom object.

1. Login to Salesforce

Open:

<https://login.salesforce.com>

Login with your Salesforce Developer Account.

2. Open Setup

1. Click Gear Icon (⚙)
2. Click:

Setup

3. Create Custom Object

Go to:

Setup → Object Manager → Create → Custom Object

Fill Object Details

Field	Value
Label	Student Record

Field	Value
Plural Label	Student Records
Object Name	Student_Record
Record Name	Student Name

Select:

- Allow Reports
- Allow Activities

Click:

Save

4. Create Custom Fields

Open:

Object Manager → Student Record → Fields & Relationships

Now create the following fields.

A. Create Roll No Field

1. Click:

New

1. Select:

Number

1. Click:

Next

1. Fill Details:

Property	Value
Field Label	Roll No
Length	5
Decimal Places	0

1. Click:

Next → Save

API Name:

Roll_No__c

B. Create Class Field

1. Click:

New

1. Select:

Text

1. Click:

Next

1. Fill Details:

Property	Value
Field Label	Class
Length	20

1. Click:

Next → Save

API Name:

Class__c

C. Create Mobile No Field

1. Click:

New

1. Select:

Phone

1. Click:

Next

1. Fill Details:

Property	Value
Field Label	Mobile No

1. Click:

Next → Save

API Name:

Mobile_No__c

5. Open Developer Console

Click:

Gear Icon → Developer Console

Developer Console opens in new window.

6. Create Apex Class

Go to:

File → New → Apex Class

Class Name:

StudentVFManager

Click:

OK

7. Paste Apex Controller Code

Delete default code and paste:

```
public class StudentVFManager {

    public Student_Record__c student {get; set;}
    public List<Student_Record__c> students {get; set;}

    public StudentVFManager() {

        student = new Student_Record__c();

        students = [SELECT Id, Name, Roll_No__c, Class__c, Mobile_No__c
                     FROM Student_Record__c];
    }

    // SAVE STUDENT
    public void saveStudent() {
```

```
        insert student;

        student = new Student_Record__c();

        students = [SELECT Id, Name, Roll_No__c, Class__c, Mobile_No__c
                      FROM Student_Record__c];
    }

    // DELETE STUDENT
    public void deleteStudent() {

        String studentId = ApexPages.currentPage().getParameters().get('sid');

        Student_Record__c s = [SELECT Id
                                FROM Student_Record__c
                                WHERE Id = :studentId LIMIT 1];

        delete s;

        students = [SELECT Id, Name, Roll_No__c, Class__c, Mobile_No__c
                      FROM Student_Record__c];
    }
}
```

8. Save Apex Class

Press:

Ctrl + S

9. Create Visualforce Page

Go to:

File → New → Visualforce Page

Page Name:

StudentVFPage

Click:

OK

10. Paste Visualforce Code

Delete default code and paste:

```
<apex:page controller="StudentVFManager">

    <apex:form>

        <apex:pageBlock title="Student Record Management System">

            <apex:pageBlockSection columns="1">

                <apex:inputText value="{!student.Name}" label="Student Name"/>

                <apex:inputText value="{!student.Roll_No__c}" label="Roll No"/>

                <apex:inputText value="{!student.Class__c}" label="Class"/>

                <apex:inputText value="{!student.Mobile_No__c}" label="Mobile
No" />

            </apex:pageBlockSection>

            <apex:commandButton value="Save Student"
                                action="{!saveStudent}"/>

        </apex:pageBlock>

    <br/>

    <apex:pageBlock title="Student Records">

        <apex:pageBlockTable value="{!students}" var="s">

            <apex:column value="{!s.Name}" headerValue="Student Name"/>
```

```
<apex:column value="{!s.Roll_No__c}" headerValue="Roll No"/>

<apex:column value="{!s.Class__c}" headerValue="Class"/>

<apex:column value="{!s.Mobile_No__c}" headerValue="Mobile No"/>

<apex:column headerValue="Delete">

    <apex:commandLink value="Delete"
        action="{!deleteStudent}">

        <apex:param name="sid" value="{!s.Id}" />

    </apex:commandLink>

</apex:column>

</apex:pageBlockTable>

</apex:pageBlock>

</apex:form>

</apex:page>
```

11. Save Visualforce Page

Press:

Ctrl + S

12. Run Visualforce Page

Go to:

Preview → StudentVFPage

OR open URL:

`https://yourInstance.salesforce.com/apex/StudentVFPage`

13. Add Student Record

Enter:

- Student Name
- Roll No
- Class
- Mobile No

Click:

Save Student

14. View Student Records

All student records will display in table format.

15. Delete Student Record

Click:

Delete

beside any student record.

16. Output

The application allows:

- Add Student Records
- View Student Records
- Delete Student Records

using Apex and Visualforce page.

17. Important Concepts

Apex

A programming language used in Salesforce for backend development.

Visualforce Page

Visualforce is a framework used to create custom user interfaces in Salesforce.

Custom Object

A custom database table created in Salesforce.

18. Conclusion

The Student Record Management System was successfully implemented using Apex and Visualforce in Salesforce. The application manages student records containing Student Name, Roll No, Class, and Mobile No stored in a custom object. The system performs add, view, and delete operations through a Visualforce user interface.